

Frontend Interface Documentation – PatternGuard

This document describes the frontend interface developed for the Aho-Corasick based PatternGuard application. The frontend allows users to add keywords, input text for scanning, and view highlighted matched patterns.

1. Initial Frontend Interface

The initial interface contains:

- Keyword input section
- Text input area
- Results display panel
- Quick preset buttons
- API status indicator

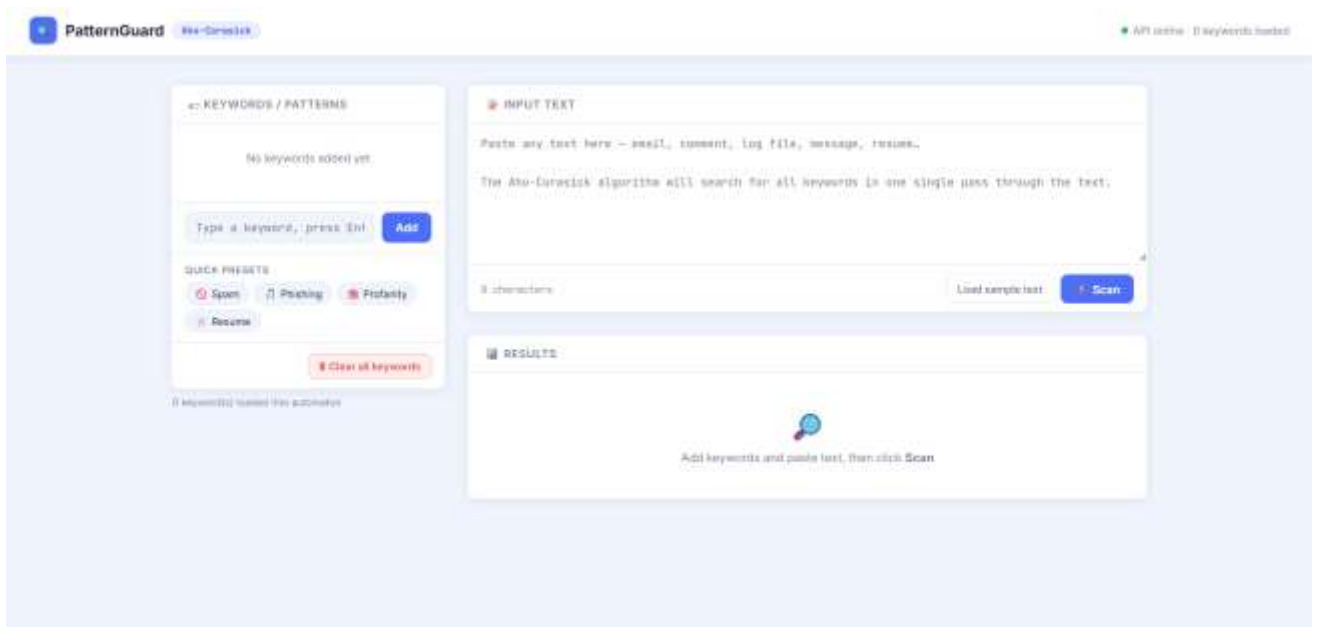


Figure 1: Initial PatternGuard Frontend Interface

2. Keyword Input Section

Users can dynamically add multiple keywords/patterns into the system. Preset categories such as Spam, Phishing, Profanity, and Resume are also available for quick testing.

Added Keywords in Screenshot:

- spam
- scam
- fraud

- free money
- winner
- prize
- act now
- click here
- limited offer
- make money fast

Figure 2: Keywords Added into the System

3. Text Input Area

The text input area allows users to paste or type text that needs to be analyzed using the Aho-Corasick algorithm. The algorithm scans the text in a single pass and detects all matching patterns.

Example Input:

A spam/phishing-style message containing suspicious keywords.

4. Results Display Section

The results section displays:

- Total matches found
- Number of keywords hit
- Characters scanned
- Scan time
- Highlighted matched patterns inside the text
- Position of each detected keyword



Figure 3: Results Display with Highlighted Keywords

5. Frontend Features

- Responsive user interface
- Dynamic keyword addition
- Quick preset categories
- Real-time text scanning
- Highlighted keyword detection

- Position tracking for matches
- API connectivity status display
- Clean and modern UI design